AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Chemistry Paper I

Time allowed: 35 minutes Marks 25

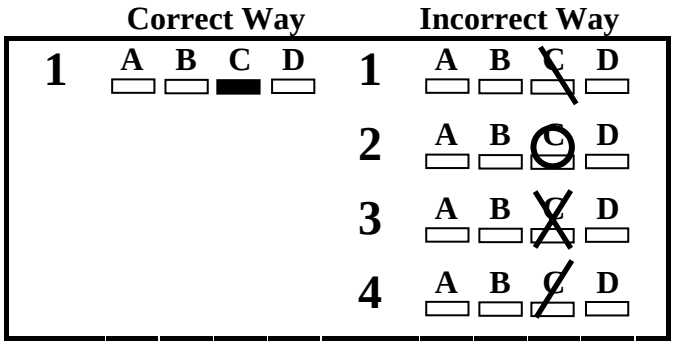
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. The equilibrium constant of a reaction is

   A. $Kc = \frac{[Reactant]}{[Product]}$

   B. $Kc = \frac{[Reactant]^2}{[Product]}$

   C. $Kc = \frac{[Product]}{[Reactant]}$

   D. $Kc = [Product] \times [Reactant]$

2. A dynamic equilibrium is the result of two opposing processes taking place at

   A. the same rate.
   B. different rates.
   C. a very fast rate.
   D. a very slow rate.

3. A solution with a pH 3 is

   A. strongly acidic.
   B. weakly acidic.
   C. strongly basic.
   D. weakly basic.

4. Which of the following is classified as a neutralization reaction?

   A. $SO_3 + H_2O \rightarrow H_2SO_4$

   B. $ZnCO_3 \rightarrow ZnO + CO_2$

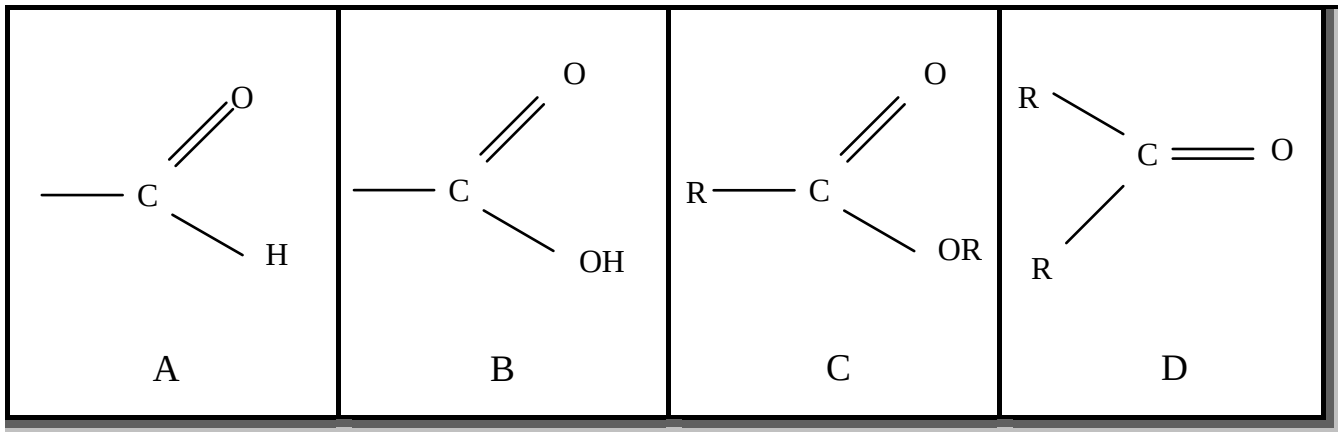
   C. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

   D. $HCl + KOH \rightarrow KCl + H_2O$

5. Which of the following is a saturated hydrocarbon?

   A. Propene
   B. Propyne
   C. Propane
   D. Propanoic acid

6. Which of the following is classified as ketone?



7. Which of the following factors is responsible for the slow rate of reaction in organic compounds?

   A. Ionic bond
   B. Covalent bond
   C. High melting point
   D. Solubility in polar solvent

8. All of the following are the characteristics of unsaturated hydrocarbons EXCEPT

   A. they are unstable.
   B. they are reactive.
   C. they contain C=C.
   D. they are known as paraffins.

9. Which of the following reacts with bromine to give tetrabromoethane?

   A. Ethane
   B. Ethene
   C. Ethyne
   D. Ethyl alcohol

10. Which of the following products are formed when the given reaction takes place?

$$C_3H_{8(g)} + 5O_{2(g)} \rightarrow ?$$

   A. $3CO_2 + 4H_2O$
   B. $3CO + 4H_2O$
   C. $3CH_4 + 6H_2O$
   D. $CH_3COOH + H_2$

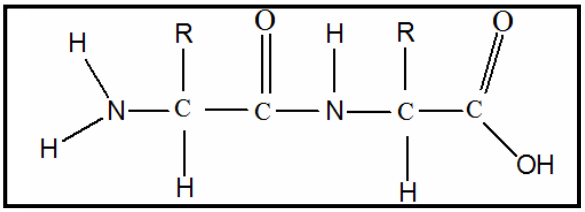
11. Which of the given characteristics of methane is considered important when it is used as a fuel?

A. It is a gas.
B. It is odourless.
C. It has a low boiling point.
D. It burns exothermically.

12. Which of the following is burnt in air to produce a smoky luminous oxy-acetylene flame?

A. Methane
B. Butyne
C. Ethyne
D. Ethanol

13. Which of the following nutrients possesses the given structure?



A. Carbohydrates
B. Fats
C. Proteins
D. Vitamins

14. Which of the following are NOT characterised as lipids?

A. Starches
B. Steroids
C. Waxes
D. Fats and oils

15. All of the following are macro-nutrients EXCEPT

A. nitrogen.
B. phosphorous.
C. potassium.
D. zinc.

16. Crude oil is purified by fractional distillation. This depends on the difference in its

A. colour.
B. density.
C. boiling point.
D. solubility.

17. In the electrolytic manufacture of aluminium, the anode (positive electrode) is made of

A. copper.
B. graphite.
C. platinum.
D. steel.

18. All of the following are alloys EXCEPT

A. brass.
B. bronze.
C. nichrome.
D. limonite.

19. The major impurity present in iron is

A. carbon.
B. cobalt.
C. magnesium.
D. sulphur.

20. Which region of the atmosphere has the maximum concentration of ozone?

A. Stratosphere
B. Hydrosphere
C. Lithosphere
D. Biosphere

21. Which of the following toxic gases is released by the oxidation of methane in the atmosphere?

A. Carbon dioxide
B. Carbon monoxide
C. Hydrogen chloride
D. Hydrogen cyanide

22. Chlorofluorocarbons rise to the stratosphere to

A. react directly with stratospheric gases.
B. interact with ultraviolet rays to destroy ozone.
C. become free radical to form ozone.
D. become free radical to remove CO_2.

23. Which of the following products is formed when water is decomposed at $2000°C$?

A. $2H_2 + O_2$
B. $H^+ + OH^-$
C. $H_3O^+ + H^+$
D. $H_3O^+ + OH^-$

24. Thermal pollution occurs at water sources near the

A. farms of animal breeding.
B. place of coal-burning.
C. plant of nuclear reactors.
D. plant of fertilizers.

25. Which of the following is responsible for the permanent hardness of water?

A. Bicarbonates of calcium
B. Bicarbonates of magnesium
C. Carbonates of calcium
D. Sulphates of magnesium

END OF PAPER

Please use this page for rough work

Please use this page for rough work